Bank Customer Churn Prediction Using Machine Learning

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ABSTRACT

Customer churn is one of the major challenges faced by modern banking institutions. Retaining existing customers is significantly more cost-effective than acquiring new ones. This project aims to develop a machine learning-based churn prediction system that identifies customers who are likely to leave the bank. Using customer demographic, financial, and engagement-related attributes, multiple machine learning models were trained and evaluated. The final solution includes a predictive model, churn probability scoring system, feature importance analysis, Power BI dashboard, and Streamlit web application for real-time churn prediction.

Keywords: Customer Churn, Machine Learning, Banking Analytics, Random Forest, Predictive Analytics, Customer Retention

1. INTRODUCTION

Customer churn refers to the loss of customers who discontinue their relationship with a bank. High churn rates negatively affect revenue, customer lifetime value, and long-term profitability.

Traditional churn analysis focuses on understanding why customers leave after the event occurs. Modern banking requires proactive identification of at-risk customers before churn occurs. This project addresses that challenge by building a predictive churn intelligence system capable of estimating churn probability and identifying important churn drivers.

2. PROBLEM STATEMENT

Banks possess large volumes of customer-level data but often lack:

- Accurate churn prediction models
- Quantitative churn risk scoring systems
- Explainable insights into churn behavior

As a result, retention efforts become reactive, inefficient, and costly.

The objective of this project is to predict customer churn using machine learning techniques and provide actionable insights for customer retention strategies.

3. OBJECTIVES

- **Primary Objectives**
- Predict customer churn with high accuracy
- Generate churn probability scores
- Identify key churn drivers

- **Secondary Objectives**
- Improve customer retention decision-making
- Enable targeted marketing campaigns
- Provide explainable machine learning insights
- Develop an interactive prediction application

4. LITERATURE REVIEW

Customer churn prediction has become an important application of machine learning in the banking sector. Previous studies have shown that customer demographics, account activity, product usage, and financial behavior significantly influence churn decisions.

Traditional statistical approaches such as Logistic Regression provide interpretable results but often struggle with complex customer behavior patterns. Ensemble methods such as Random Forest and Gradient Boosting have demonstrated superior predictive performance due to their ability to capture nonlinear relationships among variables.

Recent research highlights the importance of combining predictive analytics with explainability techniques to support business decision-making and regulatory compliance.

5. DATASET DESCRIPTION

The dataset consists of customer-level banking information collected from a European retail bank.

Feature	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer location (France, Germany, Spain)
Gender	Male/Female
Age	Customer age
Tenure	Years with the bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card ownership
IsActiveMember	Customer activity indicator
EstimatedSalary	Estimated annual salary
Exited	Target variable (1 = Churn, 0 = Retained)

The dataset contains approximately 10,000 customer records.

6. DATA PREPROCESSING

The following preprocessing steps were performed:

- **Data Cleaning**

- Removed CustomerId
- Removed Surname
- Checked dataset consistency

- **Feature Encoding**

Categorical variables were converted into numerical representations using One-Hot Encoding:

- Geography_Germany
- Geography_Spain
- Gender_Male

- **Feature Scaling**

StandardScaler was applied to normalize numerical variables before model training.

7. FEATURE ENGINEERING

Several business-oriented features were created:

- **Balance Salary Ratio**

Measures financial exposure relative to customer income.

- Formula:

Balance / Estimated Salary

- **Product Density**

Measures product ownership relative to customer age.

- Formula:

Num Of Products / Age

- **Engagement Product**

Captures customer engagement intensity.

- Formula:

Is Active Member × Num of Products

- **Age Tenure**

Represents customer maturity and relationship duration.

- Formula:

Age × Tenure

These engineered features improved the predictive capability of the models.

8. ANALYTICAL METHODOLOGY

The project followed a structured machine learning workflow consisting of data preprocessing, feature engineering, model development, evaluation, and deployment.

Step 1: Data Collection

Customer-level banking data was collected and prepared for analysis.

Step 2: Data Preprocessing

- Removed non-informative attributes
- Encoded categorical variables
- Standardized numerical features

Step 3: Feature Engineering

New behavioral and engagement features were created to improve model performance.

Step 4: Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Stratified sampling was applied to preserve churn class distribution.

Step 5: Model Development

Three machine learning models were implemented:

- **Logistic Regression**

Used as a baseline interpretable model.

- **Decision Tree**

Used to capture non-linear relationships and decision boundaries.

- **Random Forest**

Used as the primary predictive model due to its robustness and ability to reduce overfitting.

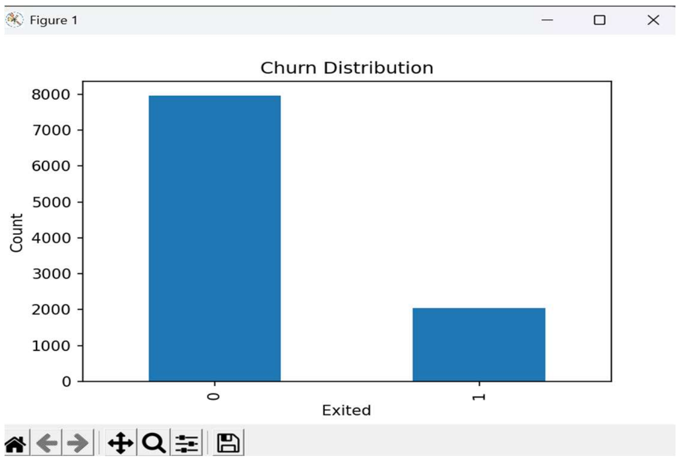
Step 6: Model Evaluation

Models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

9. EXPERIMENTAL RESULTS

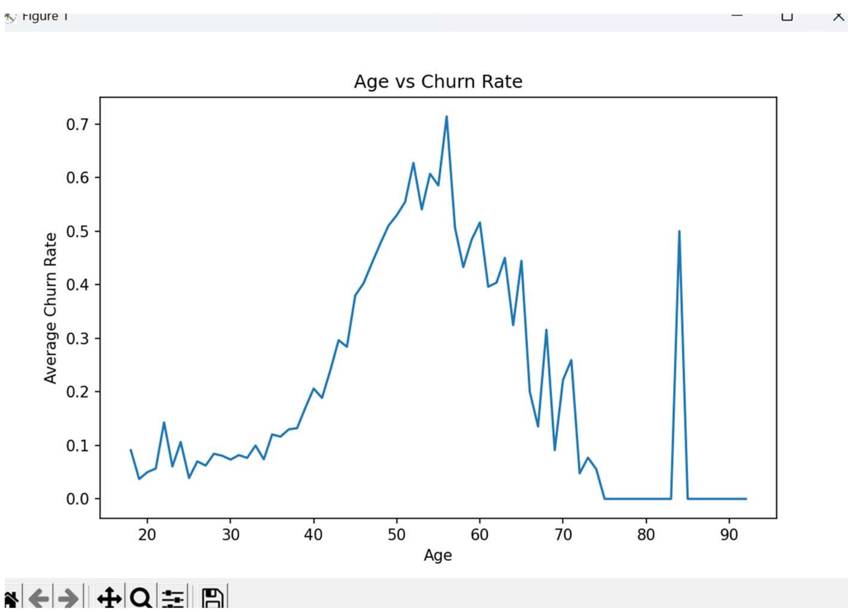
9.1 Churn Distribution:



- **Analysis:**

The majority of customers were retained, while approximately 20% customers churned. This indicates a class imbalance problem commonly observed in banking datasets.

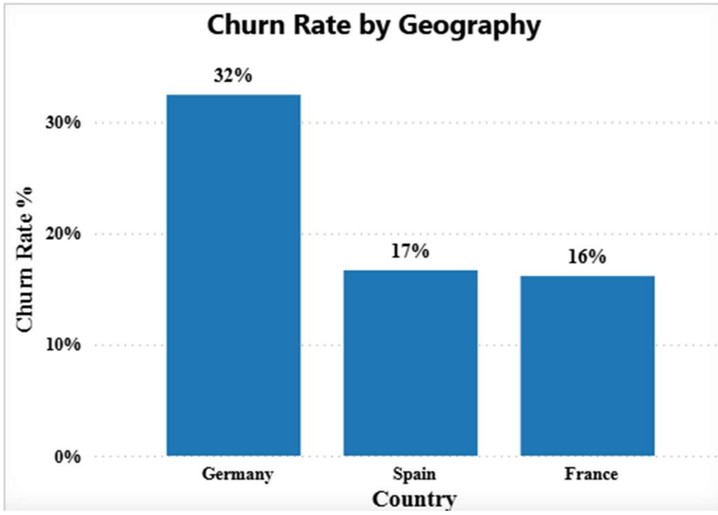
9.2 Age vs Churn Rate:



- Analysis:

Churn rate increases with age. Customers above middle-age groups exhibit a higher probability of leaving the bank.

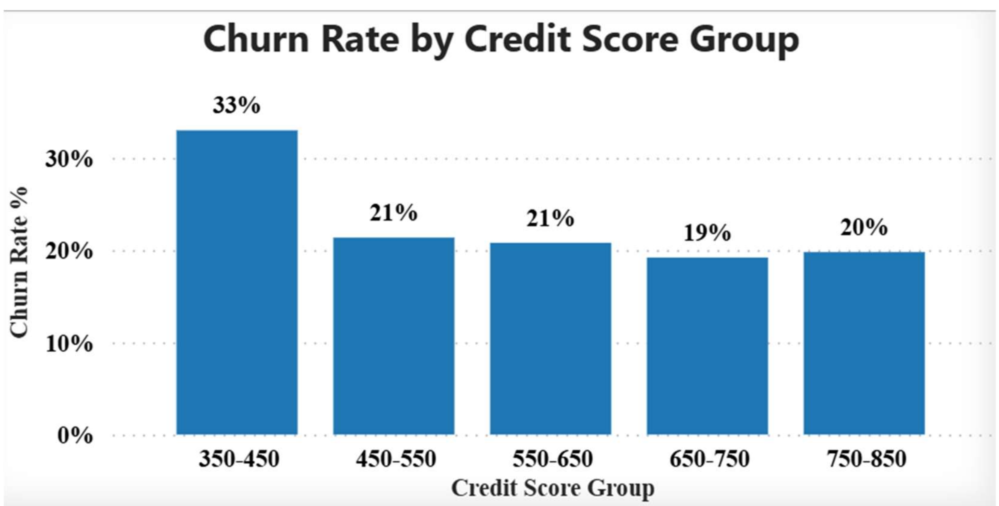
9.3 Geography vs Churn Rate:



- **Analysis**:

Germany shows a comparatively higher churn rate than France and Spain, indicating region-specific customer behavior

9.4 Credit Score vs Churn Rate:



- **Analysis:**

Customers with lower credit scores are more likely to churn compared to customers with higher credit scores.

9.5 Model Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	80.95%	0.7806
Decision Tree	79.00%	0.6834
Random Forest	86.35%	0.8453

- **Analysis**:

Random Forest achieved the highest Accuracy and ROC-AUC score, making it the best-performing model for customer churn prediction.

➢ **Logistic Regression**

- Accuracy = 80.95%
- ROC-AUC = 0.7806

➢ **Decision Tree**

- Accuracy = 79.00%
- ROC-AUC = 0.6834

➢ **Random Forest**

- Accuracy = 86.35%
- ROC-AUC = 0.8453

10. FEATURE IMPORTANCE ANALYSIS

Random Forest feature importance analysis identified the following key churn drivers:

Feature	Importance
Product Density	15.07%
Age	12.81%
Credit Score	9.99%
Estimated Salary	9.95%
Balance	9.50%

➢ **Key Findings**

- ProductDensity emerged as the strongest churn predictor.
- Age significantly influences customer retention behavior.
- Customers with lower credit scores exhibit higher churn risk.
- Salary and account balance contribute substantially to churn prediction.
- Engagement-related variables improve model performance.

11. STREAMLIT WEB APPLICATION

An interactive Streamlit application was developed to operationalize the churn prediction model.

- **Features**
- Customer churn risk calculator
- Churn probability score generation
- Risk level classification
- Probability distribution visualization
- Feature importance dashboard
- **User Functionality**

Users can enter customer information and instantly receive:

- Churn Probability (%)
- Risk Category (Low, Medium, High)
- Stay/Leave Prediction

12. POWER BI DASHBOARD

A comprehensive Power BI dashboard was created to support business intelligence and customer retention decision-making.

Dashboard Components

- KPI Cards
- Churn Rate Analysis
- Geography-Based Churn Analysis
- Age Group Analysis
- Credit Score Analysis
- Balance and Salary Segmentation
- Customer Profile Insights

The dashboard provides an executive-level overview of customer churn behavior and key risk factors.

13. CONCLUSION

Customer churn prediction is a critical business problem in the banking industry. This project successfully developed a machine learning-based churn intelligence system capable of identifying customers at risk of leaving the bank.

Three machine learning models were evaluated: Logistic Regression, Decision Tree, and Random Forest. Among them, the Random Forest model achieved the best predictive performance with an accuracy of 86.35% and a ROC-AUC score of 0.8453.

Feature importance analysis revealed that Product Density, Age, Credit Score, Estimated Salary, and Account Balance are the most influential factors affecting customer churn.

The developed Streamlit application and Power BI dashboard provide an effective decision-support framework for customer retention management by enabling proactive intervention strategies.

14. BUSINESS RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

- **Customer Engagement Enhancement**

Customers with low product density should be targeted through cross-selling and personalized product recommendations.

- **High-Risk Customer Monitoring**

Customers with high churn probabilities should be monitored through dedicated retention programs.

- **Personalized Retention Campaigns**

Retention strategies should focus on customer-specific financial and behavioral characteristics rather than broad demographic segmentation.

- **Loyalty Programs**

Long-term customers should be rewarded through loyalty programs and exclusive benefits to strengthen customer relationships.

- **Active Membership Promotion**

Inactive customers should be encouraged to increase engagement through digital banking incentives and personalized communication.

15. FUTURE SCOPE

Future enhancements can further improve the effectiveness of the churn prediction system.

- **Advanced Machine Learning Models**
- XGBoost
- LightGBM
- CatBoost
- **Explainable AI Integration**

Implementation of:

- SHAP Analysis
- Partial Dependence Plots
- LIME Explanations
- **Real-Time Prediction System**

Integration with banking systems to generate real-time churn risk alerts.

- **Customer Segmentation**

Combining churn prediction with customer segmentation for more personalized retention strategies.

- **Cloud Deployment**

Deployment on cloud platforms such as AWS, Azure, or Google Cloud for enterprise-scale implementation.

16. REFERENCES

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